



# 1. Introduction

This homework aimed to develop and analyze control systems for a quadrotor, focusing on **4 Degrees of Freedom (4DOF)** and **6 Degrees of Freedom (6DOF)** frameworks. The 4DOF system provides basic altitude and planar motion control, while the 6DOF system enhances control by adding direct manipulation of X- and Y-axes through additional forces. Simulink was used to model these systems based on the dynamical equations provided in the assignment.

The quadrotor dynamics were modeled using:

- **Translational Dynamics Equations** (in the inertial Earth frame):

$$\begin{aligned}\ddot{X} &= \frac{1}{m} (U_1 \cdot (\sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi)) , \\ \ddot{Y} &= \frac{1}{m} (U_1 \cdot (-\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi)) , \\ \ddot{Z} &= \frac{1}{m} (U_1 \cdot (\cos \theta \cos \phi) - mg) .\end{aligned}$$

- **Rotational Dynamics Equations** (in the body frame):

$$\begin{aligned}\dot{p} &= \frac{I_{yy} - I_{zz}}{I_{xx}} \cdot q \cdot r + \frac{1}{I_{xx}} U_2 , \\ \dot{q} &= \frac{I_{zz} - I_{xx}}{I_{yy}} \cdot p \cdot r + \frac{1}{I_{yy}} U_3 , \\ \dot{r} &= \frac{I_{xx} - I_{yy}}{I_{zz}} \cdot p \cdot q + \frac{1}{I_{zz}} U_4 .\end{aligned}$$

The Euler angles ( $\phi, \theta, \psi$ ) are calculated using angular velocities ( $p, q, r$ ).

The lab's goals included:

1. **4DOF**: Simulate hovering, disturbance rejection, and planar motion control using indirect roll ( $\phi$ ) and pitch ( $\theta$ ) manipulation.
2. **6DOF**: Implement trajectory tracking with takeoff, straight-line motion, circular loops, and landing.

## 2. Procedure

### 2.1. 4DOF System

The 4DOF system (figure 1) was modeled in Simulink using three primary subsystems:

#### 1. Translational Dynamics Subsystem (Figure 2):

- The translational dynamics equations were implemented using control input  $U_1$  (thrust) and Euler angles  $(\phi, \theta, \psi)$ .
- Blocks used:
  - **Sine and Cosine** functions to compute angular terms like  $\sin(\psi), \cos(\theta)$ .
  - **Gain Blocks** for terms like  $1/m$ , representing mass.
  - **Integrator Blocks** for position derivation.
- Outputs: X,Y,Z representing quadrotor positions.

For example, the Z-axis dynamics were implemented as:

$$\ddot{Z} = \frac{1}{m} \cdot (U_1 \cdot \cos \theta \cos \phi) - g,$$

integrated twice to get Z.

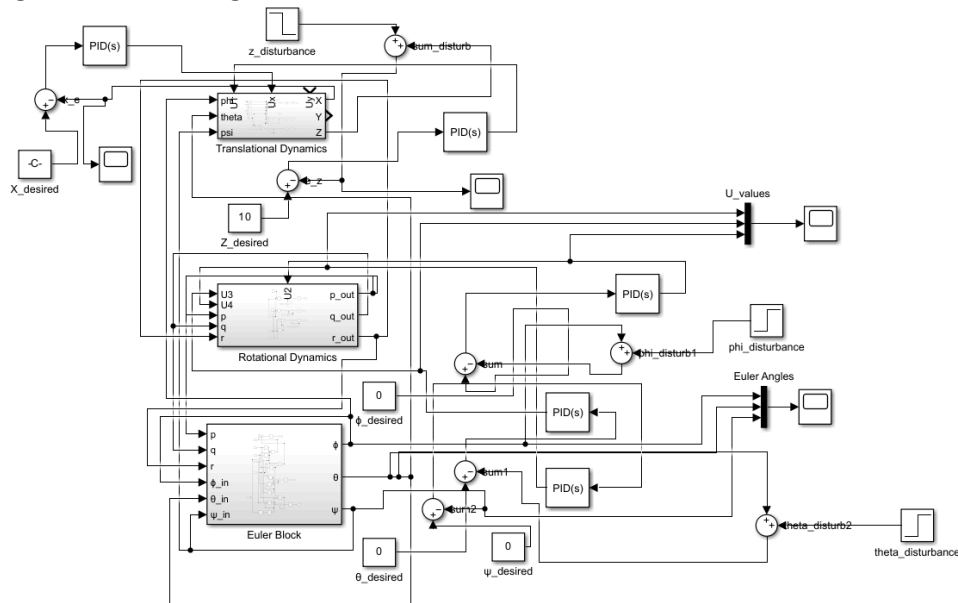


Figure 1: Main 4DOF System

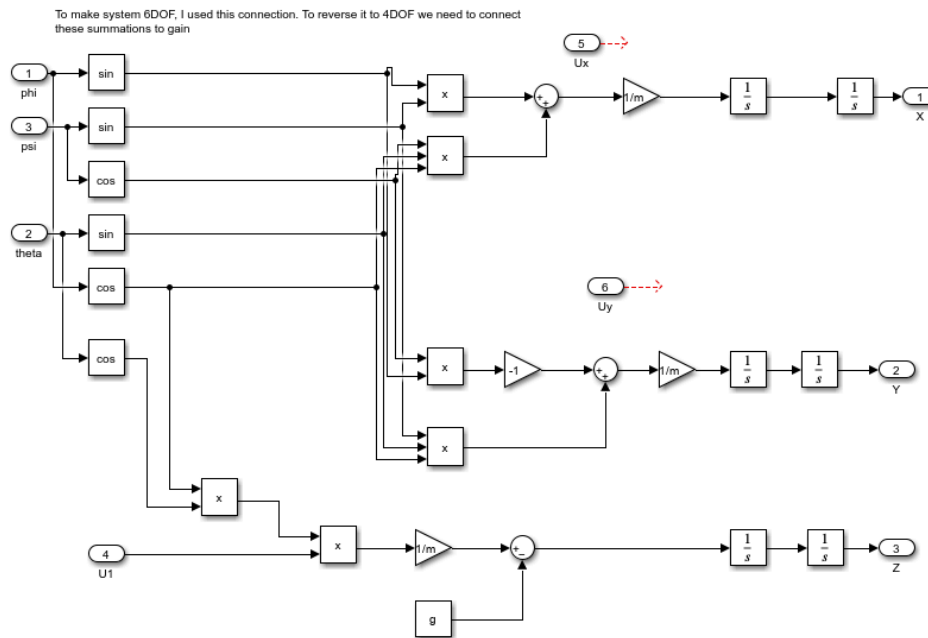


Figure 2: Translational Dynamics Subsystem of 4DOF System

## 2. Rotational Dynamics Subsystem (Figure 3):

- Rotational dynamics equations were implemented for angular velocities (p,q,r) using inputs U2,U3,U4 (control torques).
- Blocks used:

- **Product Blocks** for terms like

$$\frac{I_{yy} - I_{zz}}{I_{xx}} \cdot q \cdot r.$$

- **Gain Blocks** for inertia terms like  $1/I_{xx}$ .

- **Integrator Blocks** for angular velocities integration.

- Outputs: p,q,r representing angular velocities.

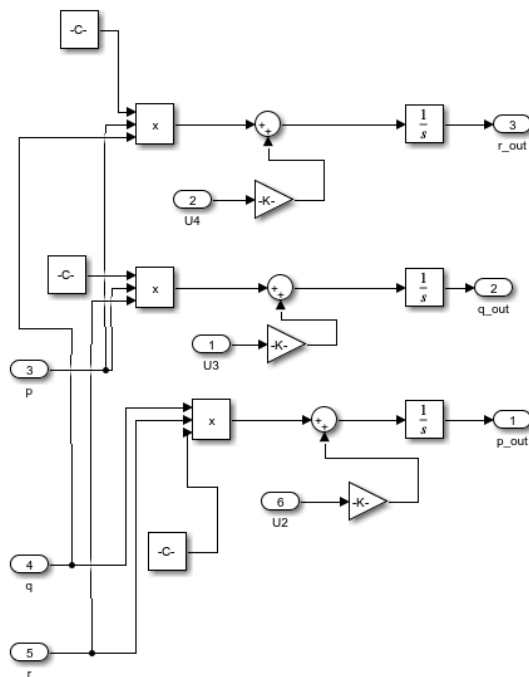


Figure 3: Rotational Dynamics Subsystem of 4DOF System

### 3. Euler Angle Subsystem (Figure 4):

- This subsystem calculated the Euler angles ( $\phi, \theta, \psi$ ) from angular velocities using the relationships:

$$\dot{\phi} = p + q \cdot \sin \phi \tan \theta + r \cdot \cos \phi \tan \theta,$$

implemented using:

- **Trigonometric Function Blocks** for sine, cosine, and tangent terms.
- **Integrator Blocks** for angle integration.
- Outputs:  $\phi, \theta, \psi$  representing roll, pitch, and yaw angles.

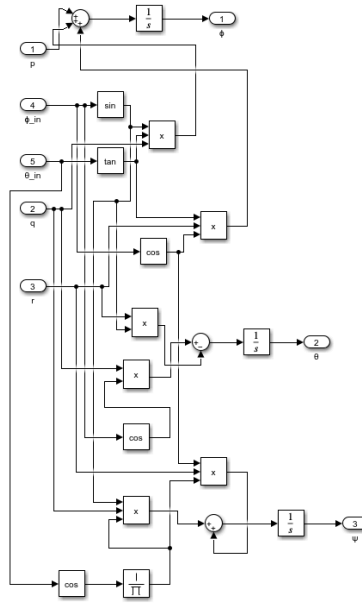


Figure 4: Euler Angles Subsystem of 4DOF System

The control structure consisted of PID controllers for Z,  $\phi$ , and  $\theta$  stabilization. Control efforts (U1,U2,U3,U4) were derived as:

- U1: Altitude control.
- U2,U3,U4: Roll, pitch, and yaw control.

## 2.2. 6DOF System

The 6DOF system (figure 5) introduced direct control of X- and Y-axes using additional control efforts Ux and Uy. The major additions were:

### 1. Translational Dynamics Subsystem (Figure 6):

- Extended to include:

$$\ddot{X} = \frac{1}{m}U_x, \quad \ddot{Y} = \frac{1}{m}U_y,$$

with  $U_x$  and  $U_y$  directly fed into the dynamics.

- Inputs:
  - $U_x$ : Horizontal force in the X-direction.
  - $U_y$ : Horizontal force in the Y-direction.
- Outputs: X,Y,Z.

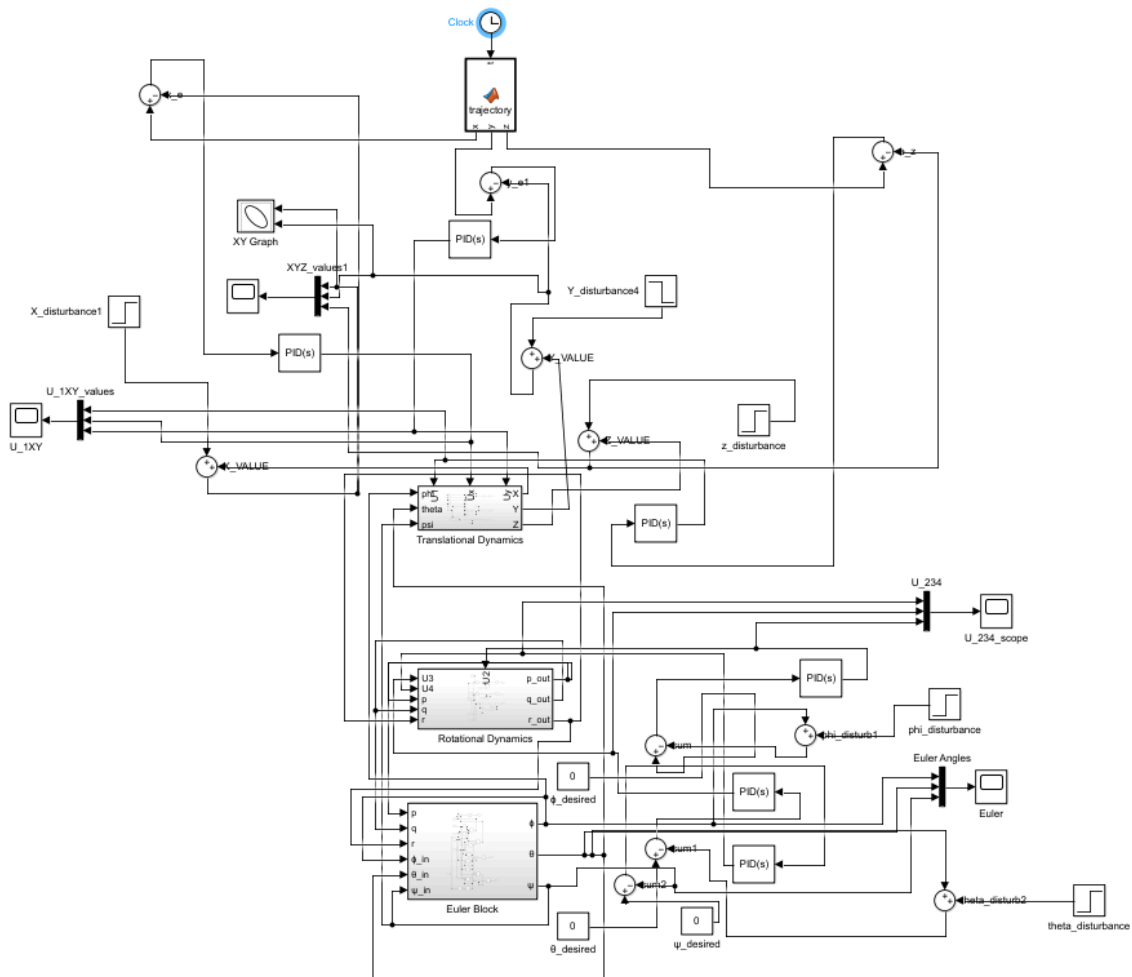


Figure 5: Main Part of 6DOF System

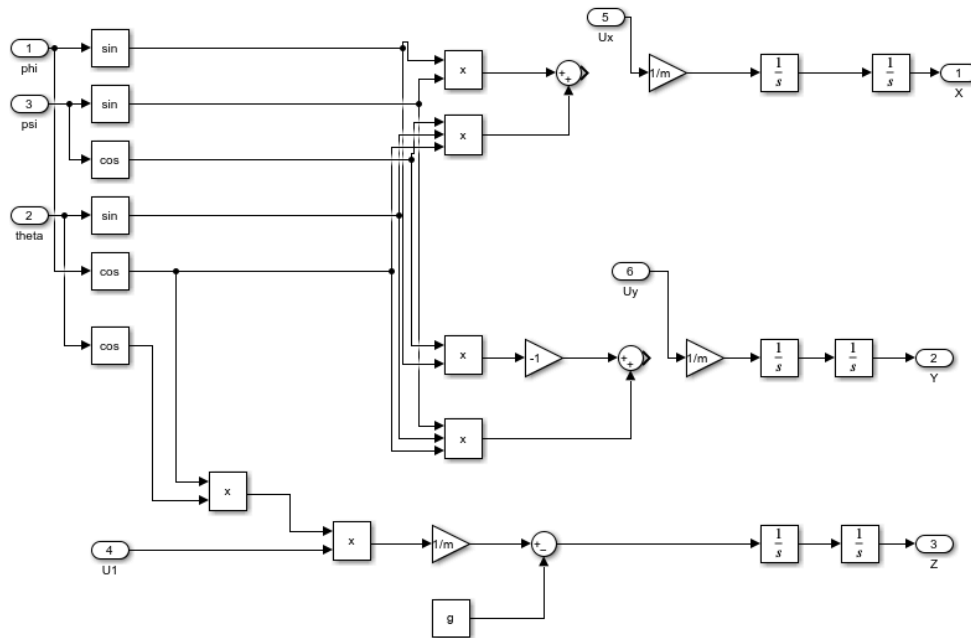


Figure 6: Translational Dynamics Subsystem of 6DOF System

## 2. Trajectory Generator (Figure 7):

- A MATLAB function defined time-based desired positions ( $X_{\text{desired}}, Y_{\text{desired}}, Z_{\text{desired}}$ ):

- Takeoff ( $t = 0$  to  $t = 5$ ):  $Z = 2 \cdot t$ .
- Straight-line motion ( $t = 10$  to  $t = 15$ ):  $X = 2 \cdot (t - 10)$ .
- Circular loop ( $t = 15$  to  $t = 25$ ):

$$X = 10 + r \cos(\omega t), Y = r \sin(\omega t), \text{ where } r = 5, \omega = 2\pi/10.$$

- Landing ( $t = 25$  to  $t = 30$ ):  $Z = 10 - 2 \cdot (t - 25)$ .



```

function [x, y, z] = trajectory(t)

    x = 0; y = 0; z = 0;

    if t >= 0 && t < 5
        z = 2 * t;

    elseif t >= 5 && t < 10
        z = 10;

    elseif t >= 10 && t < 15
        x = 2 * (t - 10);
        z = 10;

    elseif t >= 15 && t < 25
        r = 5;
        omega = 2 * pi / 10;
        x = 10 + r * cos(omega * (t - 15));
        y = r * sin(omega * (t - 15));
        z = 10;

    elseif t >= 25 && t <= 30
        z = 10 - 2 * (t - 25);
    end
end

```

Figure 7: Trajectory function to test 6DOF System

### 3. Control System:

- PID controllers were used for each position axis (X,Y,Z) and attitude ( $\phi, \theta, \psi$ ) to generate control efforts ( $U_x, U_y, U_1, U_2, U_3, U_4$ ). An example can be seen in figure 8.

|                   |                                |
|-------------------|--------------------------------|
| Proportional (P): | <input type="text" value="7"/> |
| Integral (I):     | <input type="text" value="3"/> |
| Derivative (D):   | <input type="text" value="5"/> |

Figure 8: PID Control of Y

Rotational dynamics (figure 3) and Euler Angle (figure 4) part is the same as 4DOF system, because they are already considered as 3 degree of freedom of the system.

# Discussion and Analysis

## 4DOF System

### Hovering at Desired Altitude with Initial Nonzero Roll and Pitch Angles

When the quadrotor starts on the ground with small nonzero roll ( $\phi$ ) and pitch ( $\theta$ ) angles, the goal was to stabilize at a desired altitude  $Z$  through the feedback control system. The analysis of the **altitude with disturbance** plot demonstrates the system's response to external forces:

#### 1. Altitude Control:

- From the **altitude with disturbance** plot (Figure 9), the quadrotor successfully stabilizes at the desired altitude after a small transient period. The disturbance at  $t = 3s$ , temporarily causes a deviation, but the PID controller compensates effectively to bring  $Z$  back to the desired value  $Z = 10$ .

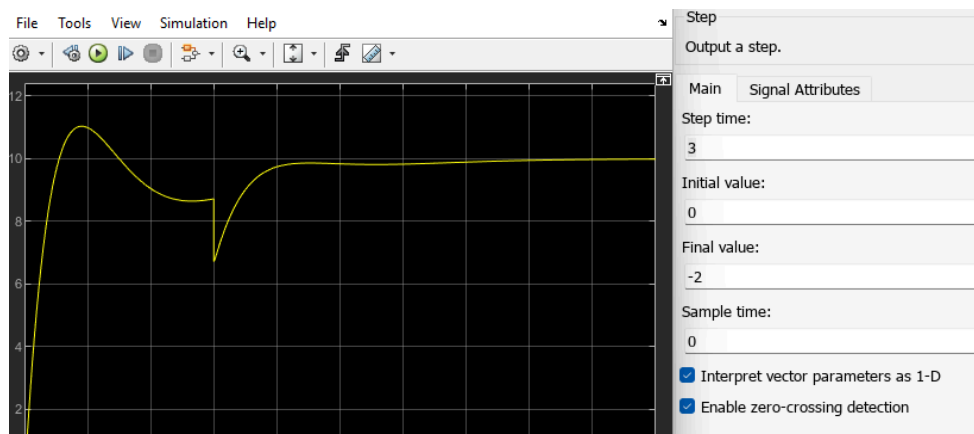


Figure 9: Altitude Plot with Disturbance

#### 2. Control Efforts (U-values):

- The plot of control efforts (Figure 10) shows that  $U_1$  dominates as expected since it directly controls thrust. The values for  $U_2, U_3, U_4$  which correspond to roll, pitch, and yaw torques, show significant spikes during stabilization,

particularly during disturbance rejection but they are handled and stabilized.

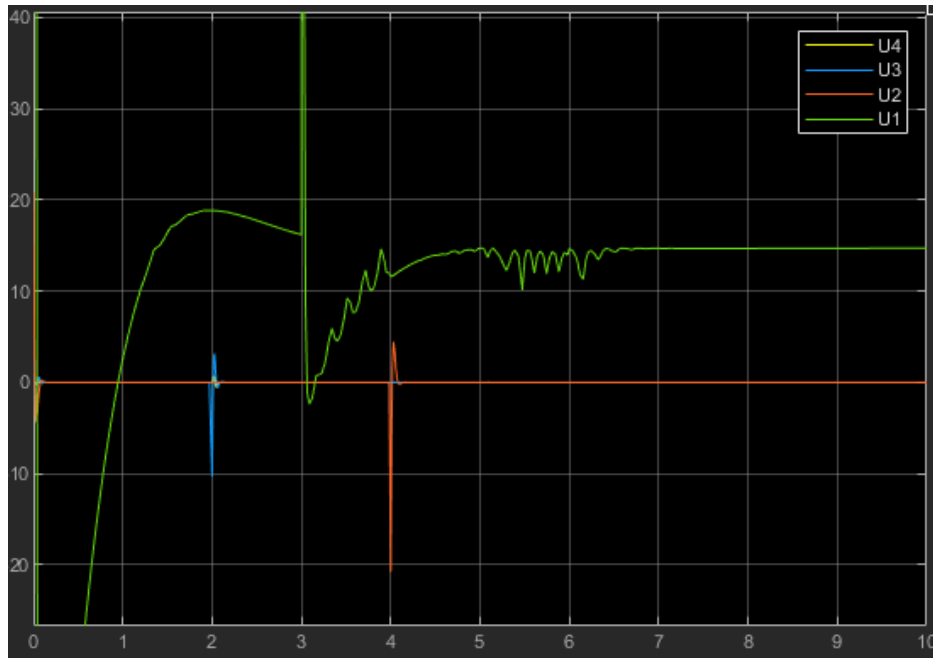


Figure 10: Plot of U values in the 4DOF System

### 3. Euler Angles:

- The **Euler angles plot** (Figure 11) illustrates how roll and pitch angles stabilize to zero over time. The transient spikes correspond to the initial nonzero angles and disturbance at  $t=3$ .
- The smooth transition back to zero validates that the rotational dynamics subsystem, paired with PID feedback control, effectively corrects angular displacements.

**Conclusion for Altitude Stability:** The quadrotor can hover at a desired altitude even with initial tilts. The control system stabilized it, in several seconds.

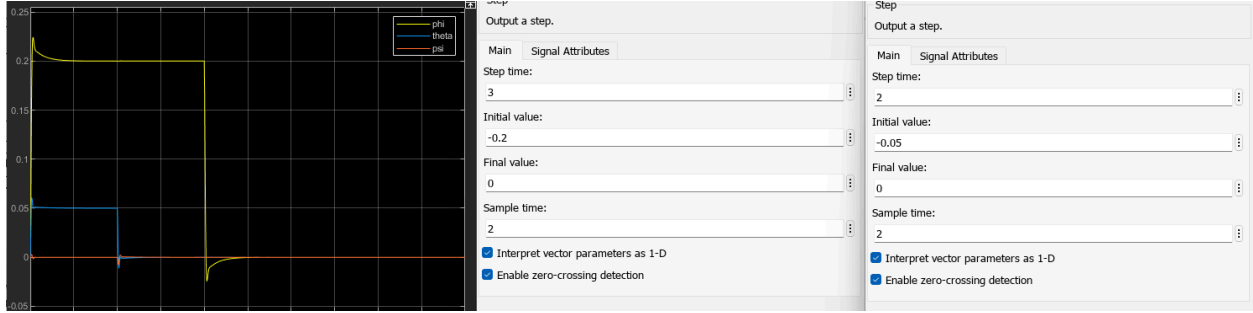


Figure 11: Euler Angles with Disturbance

## Motion in the X-Y Plane

The provided **plots (figure 12) for X-direction control** highlight a problem:

In a 4DOF system, motion in the X-axis is directly influenced by both pitch angle ( $\theta$ ) and roll angle ( $\phi$ ). The relationship arises from the translational dynamics equation for X:

$$\ddot{X} = \frac{1}{m} (U_1 \cdot (\sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi)) ,$$

Here,  $U_1$  represents the thrust force. When the pitch angle  $\theta$  deviates from zero, it causes a component of the thrust to act in the X-direction, leading to acceleration along that axis. This explains why even in a 4DOF system, where control is focused on altitude and yaw, pitch indirectly affects horizontal motion, creating X-axis displacement.

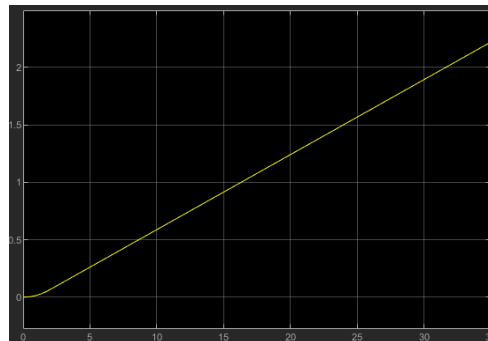


Figure 12: X motion in 4DOF System

## 6DOF System

The 6DOF (Six Degrees of Freedom) system simulation reflects a more comprehensive approach compared to the 4DOF system. This allows for the translational and rotational dynamics to be controlled independently, enabling precise movement in both position and orientation. Here's a detailed discussion of the results observed from the 6DOF plots and how they correlate with the behavior of the quadrotor:

### 1. Translational Dynamics

- **Plots: X, Y, and Z Values Over Time (figure 13):**
  - In the plot showing the trajectory of X, Y, and Z positions over time, the motion aligns with the expected trajectory provided by the trajectory(t) function.
  - The Z value increases linearly as the quadrotor ascends, stabilizes at the desired altitude during horizontal movement, and finally descends smoothly.
  - Both X and Y exhibit behavior influenced by the circular trajectory implemented between  $t = 15$  to  $t = 25$ . The sinusoidal components in this period show the quadrotor's effective tracking of the circular path, demonstrating the proper functioning of the PID controllers for translational dynamics.
- **Disturbance Rejection (figure 13):**
  - The disturbances in the X and Y directions are effectively rejected. This is evident from the stabilization of the trajectory after the disturbances are introduced. The system maintains its trajectory, proving the robustness of the PID controllers.

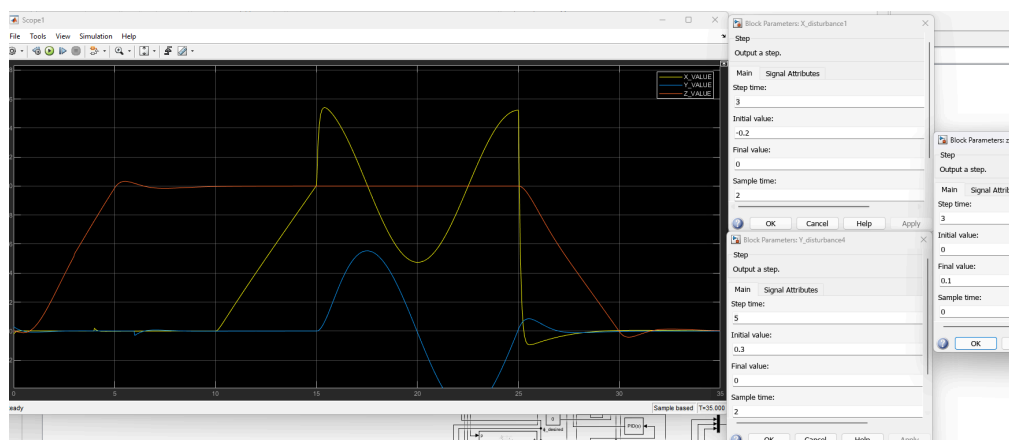


Figure 13: X, Y, Z Values  
During Test with Disturbance

## 2. Rotational Dynamics

- **Plots: Euler Angles Over Time (figure 14):**

- The Euler angles ( $\phi$ ,  $\theta$  and  $\psi$ ) show smooth transitions and stabilize over time.
- During the circular motion, there is a noticeable change in  $\phi$  and  $\theta$ , which corresponds to the roll and pitch angles required to follow the curved path.
- Despite disturbances added to roll and pitch angles, the controllers effectively bring these angles back to the desired states, indicating the robustness of rotational dynamics control.

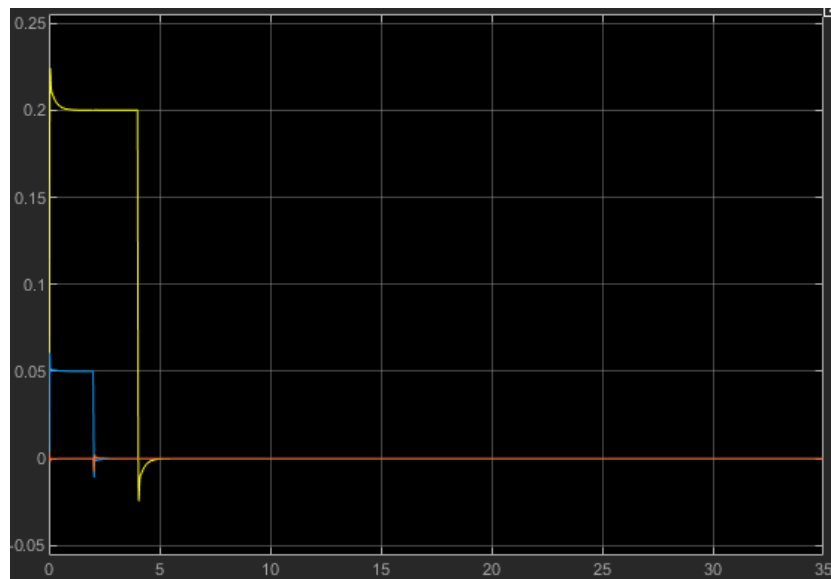


Figure 14: Euler Angle Values  
During Test with Disturbance

### 3. Control Efforts

- **Plots: Control Signals (figure 15):**

- The control signals ( $U_1$  for thrust,  $U_2$ ,  $U_3$ , and  $U_4$  for rotational dynamics) exhibit smooth variations.
- Notable spikes in  $U_x$  and  $U_y$  occur during transitions and when disturbances are introduced, which is expected behavior. These spikes represent the controllers counteracting the disturbances and ensuring stability.
- The smooth settling of these signals after disturbances confirms the system's ability to maintain stability without oscillations or excessive overshoot.

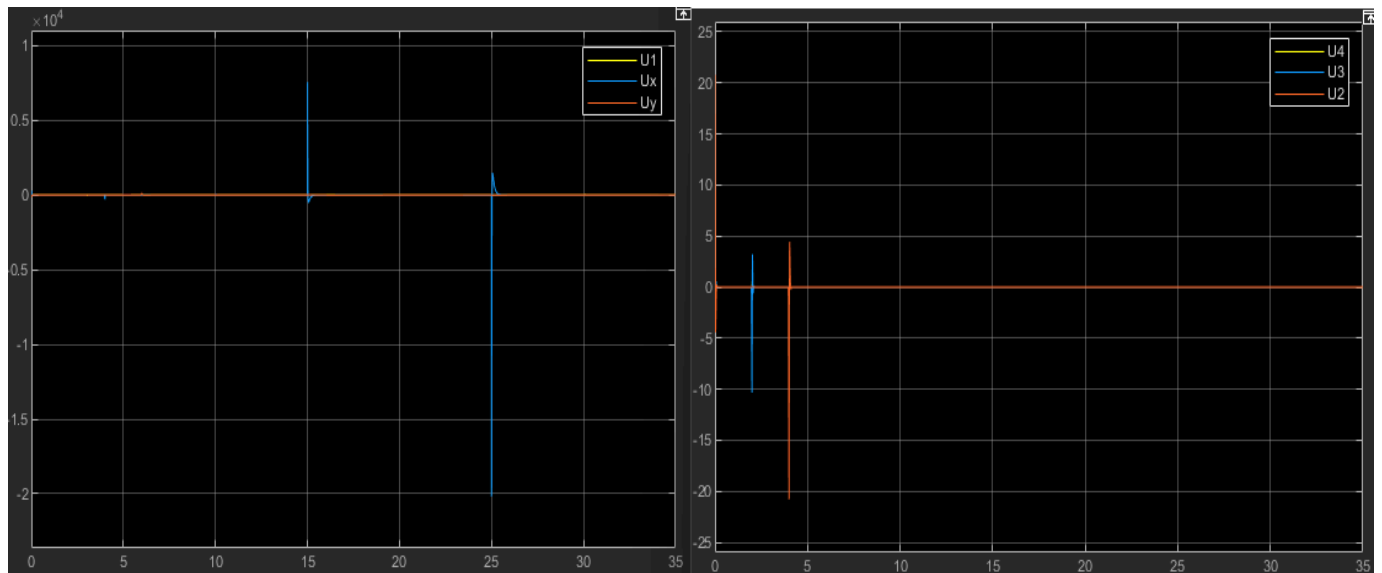


Figure 15:  $U_{1,2,3,4}$  and  $U_{x,y}$   
Values During Test with  
Disturbance

### 4. XY Plane Trajectory

- **Plot: XY Trajectory (figure 16):**

- The circular trajectory is clearly observed in this plot. The trajectory smoothly transitions into the circular path during the specified time window.

- Minimal deviations from the desired path demonstrate the effectiveness of the PID tuning for maintaining control over the X and Y positions.

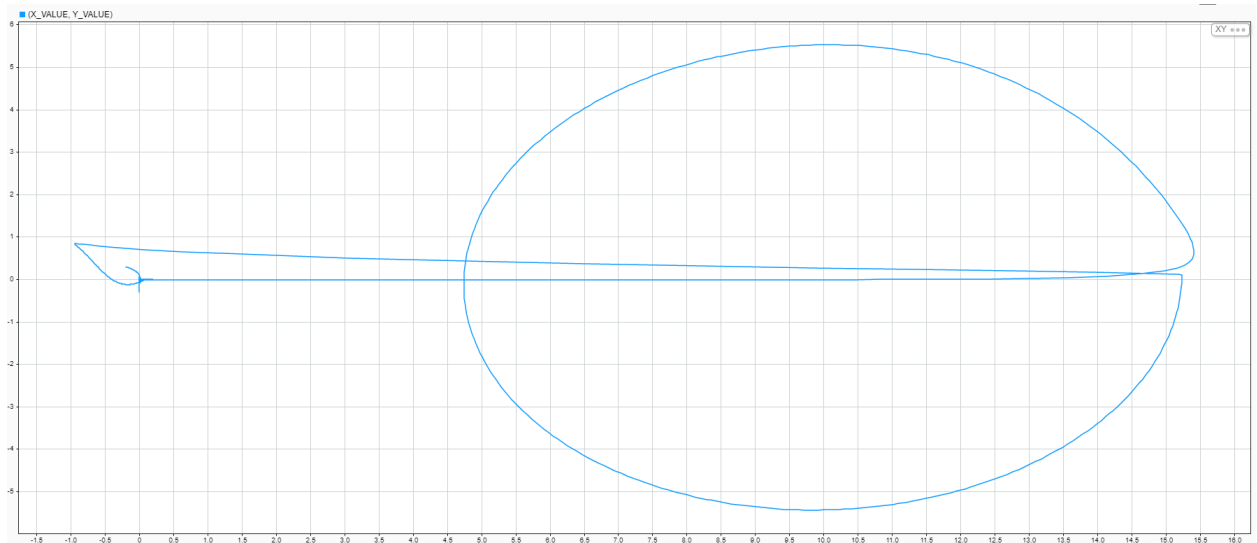


Figure 16: X-Y Trajectory of 6DOF System During Test

## Conclusion

In this lab, we successfully designed and implemented both 4DOF and 6DOF quadrotor models in Simulink, showcasing the behavior and control of a quadrotor system. The simulation demonstrated the quadrotor's ability to achieve stable flight, follow desired trajectories, and reject external disturbances through effective PID control.

The 4DOF system focused on vertical motion (altitude) and rotational stability, showing the fundamental dynamics of a quadrotor. Although it provided basic control for altitude and orientation, its ability to move in the X-Y plane, seem to be dependant on angles. Disturbances introduced in the roll or pitch angles affected the ability to hover and maintain the desired altitude, which further emphasized the need for translational and rotational dynamics.

The 6DOF system extended the capabilities of the 4DOF model by incorporating full translational and rotational control. This allowed for precise movements in all three



spatial directions and provided the ability to execute complex trajectories, such as circular motion. Disturbances introduced in translational and rotational dynamics were effectively counteracted by well-tuned PID controllers, ensuring stability and robustness of the system. The integration of trajectory generation, including takeoff, hovering, circular motion, and landing, demonstrated the quadrotor's capability to follow complex commands while maintaining stability.

Key insights gained from this simulation include:

- The importance of coupling translational and rotational dynamics for achieving full control over the quadrotor's motion.
- The role of well-tuned PID controllers in rejecting disturbances and minimizing overshoot, ensuring smooth and stable system behavior.
- The 6DOF model's superior ability to handle complex maneuvers and maintain stability under varying conditions.

Overall, this homework provided a comprehensive understanding of quadrotor dynamics and control, highlighting the challenges and intricacies of designing such systems. The results reflect the successful implementation of the models and control strategies.

## Appendix

Parameters: These are parameters that executed before running our simulation in matlab environment.

```
m = 1.5;  
g = 9.81;
```

```
Ixx = 0.02;  
Iyy = 0.02;  
Izz = 0.04;
```

```
U1_max = 20;  
U2_max = 1;  
U3_max = 1;  
U4_max = 1;
```